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المدرسة الوطنية العليا للتكنولوجيات المتقدمة

Report of Educational Visit to

CDTA

Advanced Technology Development Center

Student in 3rd year of the Engineering Cycle
Telecommunications and Networking Systems

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Acknowledgment

I would like to express my gratitude to everyone at the Centre for the Development of Advanced Technologies (CDTA) for their support during our visit. Their willingness to share expertise and demonstrate their projects has helped deepen our understanding of advanced technologies and the challenges in this field. I also thank our school for organizing this educational visit and for its continued support of our academic and professional development.

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Chapter 1

Introduction

1.1 Visit Objectives

As part of our Telecommunications and Networking Systems engineering curriculum, we had the opportunity to visit CDTA, a pivotal institution in Algeria's technological innovation landscape. The visit allowed us to witness firsthand the cutting-edge research and interdisciplinary projects addressing both industrial and societal challenges. This report aims to synthesize our observations, detailing the operational structure of CDTA and its diverse research areas while highlighting the integration of advanced technologies such as artificial intelligence, machine learning, and state-of-the-art communications systems.

1.2 Presentation of CDTA

The Center for Advanced Technology Development (CDTA) is a key national actor in the advancement of high technology in Algeria, established in 1988 and now in its fourth decade, is a leading public research institution that drives technological innovation across diverse fields—from robotics, AI, and telecommunications to microelectronics and nanotechnology. Under the guidance of the Ministry of Higher Education and Scientific Research, CDTA focuses on developing solutions to meet industrial and societal needs. The institution adopts a multidisciplinary approach that fosters collaboration among universities, research laboratories, and industry partners. CDTA's work is organized around six strategic sectors:

- Health
- Energy
- Environment
- Water
- Digital Technologies
- Industrial Technologies

This strategic orientation not only drives national innovation but also aims to enhance Algeria's international presence in the high-tech domain.

1.2.1 Organization

The organizational structure of CDTA is designed to facilitate efficient workflow and promote clear communication across its multidisciplinary teams. Figure 2.1 displays the comprehensive organization chart, outlining the hierarchical and functional relationships within the center.

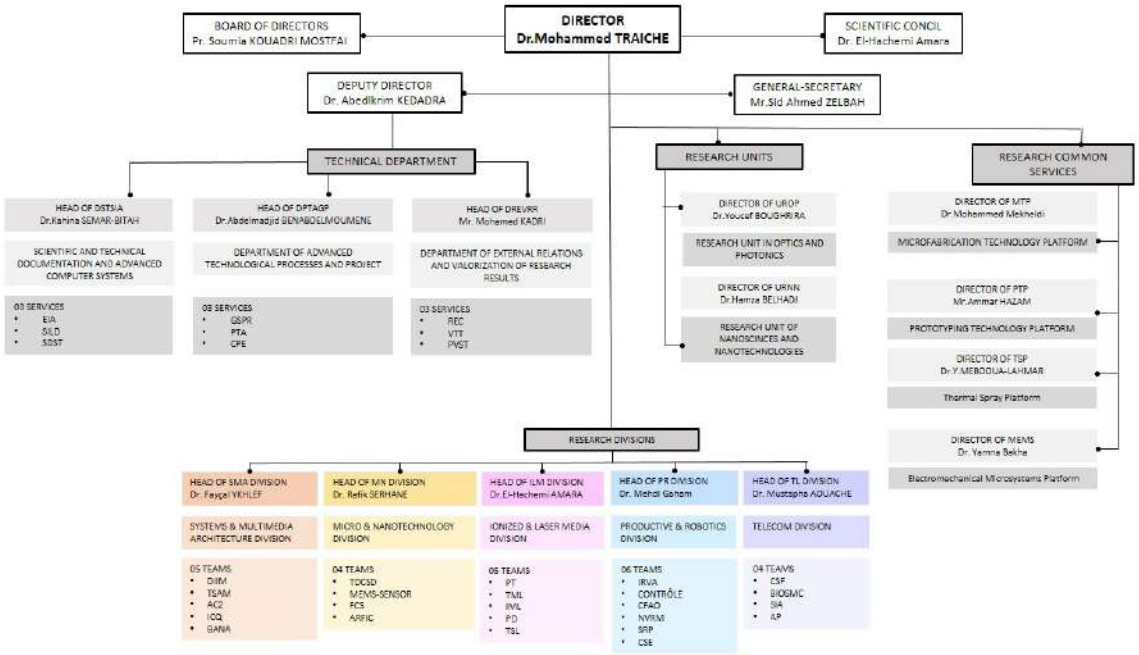

Figure 1.1: CDTA Organizational Structure

1.2.2 Vision and Missions

CDTA is committed to advancing state-of-the-art, multidisciplinary technologies through over three decades of expertise in research and development. The center continually pushes the boundaries of innovation by integrating diverse technical fields to create adaptable and evolving solutions that address national challenges.

To further this goal, CDTA is expanding its international engagement by hosting visiting researchers and entering into multilateral collaborations—adopting global R&D standards such as the TRL scale. Although its products may not always match the aesthetics of commercial offerings, they excel through:

- Tailoring solutions to specific challenges,
- Evolving continuously with technological advances,
- Providing expert guidance from initial technical specifications to full-scale implementation.

In parallel, CDTA maintains strong academic partnerships, ensuring that each researcher contributes to both training and supervision. This dual approach reinforces its mission: to put science and technology at the service of society and the citizen.

Chapter 2

Divisions

2.1 Overview

CDTA is organized into several specialized divisions that drive its multidisciplinary approach to advanced technology development. These divisions span a broad spectrum of research areas—from systems architecture and multimedia to microelectronics, nanotechnologies, ionized media, lasers, robotics, industrial automation, and telecommunications. During our visit, we had the opportunity to explore the following key divisions:

- System and Multimedia Architecture Division
- Telecom Division
- Robotics and Industrial Automation Division

Each division is dedicated to pioneering innovative solutions tailored to address specific technical challenges and national priorities.

2.2 Division of System and Multimedia Architecture (ASM)

The ASM Division is a dedicated R&D unit committed to advancing signal and image processing, biometrics, cryptography, and artificial intelligence. Its core missions include:

- Conducting high-quality research to address industry and societal challenges.
- Enhancing professional training in mathematics, computer science, and electrical engineering.
- Providing robust supervision for Bachelor, Master/Engineer, and PhD students.

Organizational Structure and Teams

The division is organized into several specialized teams:

- **Architecture for Classification and Cryptography (AC2)**
- **Signal Processing and Medical Applications (TSAM)**

- **Instrumentation and Control of Quality (ICQ)**
- **Design and Implementation of Intelligent Machines (DIIM)**
- **Binary Arithmetic for Numeric Applications (BANA)**

Research Focus and Key Applications

The ASM Division focuses on developing practical, AI-driven solutions, with notable research and applications including:

- **Medical Image Analysis:**
  - Detecting anomalies in blood images and mammograms to identify malignant cancer cells.
  - Estimating tumor volume and cancer staging via advanced 3D segmentation—providing a non-invasive alternative to traditional methods.
  - Developing semi-automated systems that use international datasets and intuitive user interfaces to support remote diagnosis, notably during the COVID-19 crisis.
- **Advanced Image Processing and Machine Learning:**
  - Intelligent License Plate Recognition: Creating a computer vision-based system for automated vehicle identification.
  - Implementing algorithms to classify and segment images where data points are not easily separable. For example, in facial recognition, machine learning techniques (such as SVM) are used to augment the feature space (e.g., by adding a z-axis with appropriate weighting) to effectively separate overlapping data clusters without resorting to overly complex polynomial models.
- **Satellite Image Analysis:**
  - Analyzing satellite images from regions such as Biskra (collected in 1990, 2000, 2006, and 2011) to monitor environmental changes, such as desertification and shifts between vegetated and sandy areas.
  - Utilizing radiometric values of each pixel to classify areas (e.g., desert vs. vegetation) and generalize the model to broader image sets.
- **Additional Applications:**
  - Processing signals from existing databases for applications such as accident analysis or assessing pain tolerance.

During our visit, we observed these innovative capabilities in action, demonstrating ASM's commitment to integrating cutting-edge research with real-world applications in sectors such as healthcare and public security.

2.3 Telecom Division

Telecom Division (DT), established in 2014, has built a strong reputation in R&D across radio frequencies, antenna design, wireless communications, advanced information systems, computer vision, multimedia security, biometrics, forensics, and cloud computing applications. Its objectives include achieving both national and international scientific positioning, exploring new research directions, and developing deep expertise in its core fields.

Missions and Strategies

- Develop innovative approaches for wireless intelligent access and mobility management in advanced telecommunication systems.
- Pioneer novel methods in biometric recognition and multimedia security.
- Advance open information systems by leveraging modern web technology standards.
- Design and implement cutting-edge RF devices for diverse applications.
- Establish a center of excellence in research covering antennas, propagation, wireless sensors, embedded systems, cloud infrastructures, and multimedia acquisition.
- Develop E-health applications and decision support systems.
- Provide high-level training and supervise final-year projects and doctoral theses.

Research Teams

The division comprises four specialized research teams:

- **Radio Frequencies and Antennas (RFA)**
- **Wireless Communications (CSF)**
- **Advanced Information Systems (SIA)**
- **Biometrics, Multimedia Security and Forensics (BIOSMC)**

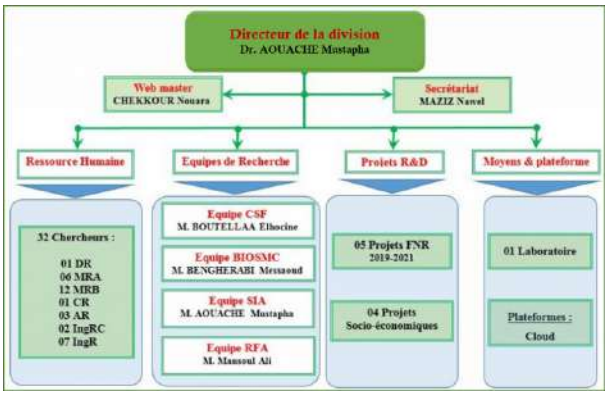

Figure 2.1: DT Organigram

DT actively fosters research collaborations with national and international partners, ensuring continuous innovation and career development for its researchers.

2.3.1 Radio Frequencies and Antennas (RFA)

The RFA team is at the forefront of designing, simulating, and optimizing RF circuits and antennas for modern wireless systems. Their work not only involves the fabrication and comprehensive study of RF circuits, but also the meticulous design and testing of antennas using advanced simulation software and state-of-the-art measurement tools. The team ensures that every design meets the high-performance standards demanded by today's telecommunications industry by rigorously evaluating key parameters such as directivity, gain, and efficiency.

Testing is carried out in highly controlled environments—most notably in our anechoic chambers where the radiation patterns and overall performance of each antenna are precisely measured. To further enhance the accuracy of these measurements, the team employs a sophisticated network vector analyzer, which is critical for the detailed characterization of RF circuits across various frequency bands. Equally important is the use of a comprehensive calibration kit, an indispensable tool that standardizes the response of measurement equipment. This kit not only ensures that the vector analyzer and other instruments are operating at peak precision, but also guarantees that all test results are both accurate and repeatable, forming the backbone of our quality control process.

Figure 2.2 shows the network vector analyzer used for precise RF circuit testing, while Figure 2.3 highlights the calibration kit that underpins the reliability of these measurements. Figure 2.4 illustrates the anechoic chamber where antenna performance is rigorously evaluated, and Figure 2.5 presents the test antennas developed through this iterative design process.

Figure 2.2: Network Vector Analyzer used for RF circuit testing

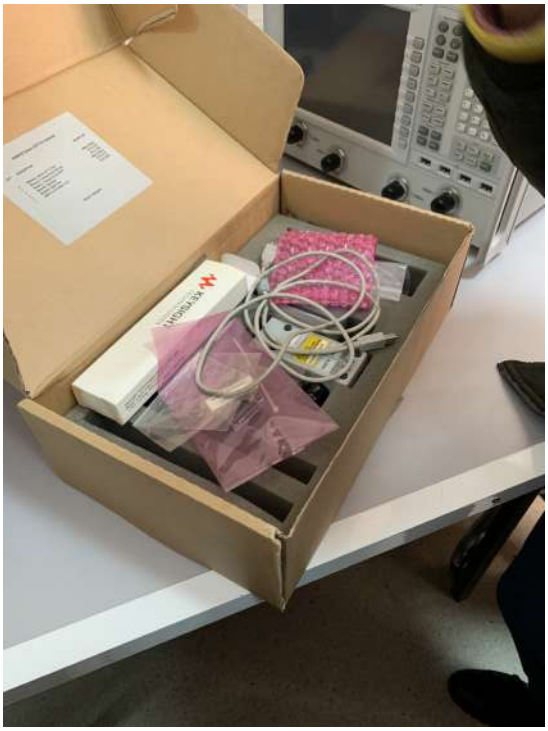

Figure 2.3: Calibration Kit for RF measurements

Figure 2.4: Anechoic Chamber for antenna performance evaluation

Figure 2.5: Test antennas used in the design validation process

2.3.2 Wireless Communications (CSF)

Dedicated to wireless network research and intelligent communication systems, the CSF team has developed several key projects:

- *Wireless Communications for Healthcare:* A wireless body area network that captures biometric signals (e.g., EEG, ECG) from cardiac patients, transmitting data via LoRa for remote monitoring and diagnostic support.
- *Antenna Selection in Massive MIMO (5G):* AI-driven algorithms dynamically optimize antenna activation in a 64×64 and more Massive MIMO system by analyzing real-time channel state information (CSI) to enhance throughput and energy efficiency.
- *Adaptive Modulation Detection (5G):* Utilization of deep learning techniques (such as CNN, RNN, and LSTM) to automatically identify and adjust modulation schemes based on channel conditions. This adaptive approach ensures reliable data transmission despite varying signal quality.

2.3.3 Advanced Information Systems (SIA)

Specializing in computer vision and advanced software solutions, the SIA team has spearheaded projects such as:

- *Intelligent Waste Management:* A system that employs computer vision to monitor bin fill levels in real time and optimize waste collection routes, thereby improving operational efficiency.

- *Medical Ocular Diagnosis:* An AI-driven application designed to assist ophthalmologists in detecting and evaluating ocular pathologies, facilitating early diagnosis and informed treatment planning.

2.3.4 Biometrics, Multimedia Security and Forensics (BIOSMC)

Focusing on biometric recognition, multimedia security, and forensic analysis, the BIOSMC team develops robust security solutions that underpin secure communications and data integrity across the division.

These projects underscore the Telecom Division's commitment to leveraging cross-disciplinary expertise—from wireless sensor networks and deep learning to RF engineering—to address both healthcare and telecommunication challenges effectively.

2.4 Robotics and Industrial Automation Division

The Robotics and Industrial Automation Division (DPR) comprises sixteen researchers and support staff dedicated to advancing modern robotics, cyber-physical systems, and production technologies. The division is active in diverse fields such as Mobile, Service, and Industrial Robotics; Manufacturing; Cyber-physical Systems; the Industrial Internet of Things; Automated Production Systems; Virtual and Augmented Reality; and Precision Engineering. Researchers are organized into specialized teams, each equipped with dedicated robotic platforms and experimental setups to develop solutions that meet socio-economic needs.

Missions and Objectives

The division pursues a broad spectrum of research activities related to production and robotics, emphasizing an experimental R&D approach that leverages:

- **Experimental Platforms:** Robotic experimentation and validation platforms, along with a flexible robotic assembly system, enable the demonstration of Industry 4.0 technologies.
- **Cyber-physical Manufacturing:** Theoretical and experimental investigations of methods and technologies for advanced robotic systems and manufacturing processes.
- **Virtual and Augmented Reality:** Development of software models and techniques to support multimodal interaction and collaboration in virtual environments.
- **Advanced Machining:** Utilization of five-axis digital milling for producing complex free-form surfaces.
- **Control Strategies:** Design of robust controllers for complex, non-linear, underactuated, and strongly coupled dynamic systems.
- **Aerial Robotics:** Study of flying vehicle dynamics (birotor, quadrotor, and plane) and the design of robust commands to mitigate control challenges.

- **Autonomous Vehicles:** Implementation of advanced features on the prototype electric vehicle, *Robucar*, for automated urban transport with intelligent navigation.
- **Robotic Guidance:** Development of a robotic guide platform incorporating logical reasoning systems to optimize human-robot interaction.
- **Manufacturing Systems:** Creation of exact, heuristic, optimization-simulation, and meta-heuristic approaches for the scheduling and control of flexible, complex manufacturing systems.
- **Interaction and Simulation:** Advanced simulations and interaction techniques for cyber-physical manufacturing and robotics systems.

Projects Observed During the Visit

During our visit, we witnessed a range of innovative projects that vividly illustrate the division's tangible impact on modern industry by merging advanced research with practical applications.

- **Embedded Systems and PCB Fabrication:** In collaboration with Sonelgaz, the division has engineered state-of-the-art smart gas meters through the development of sophisticated embedded systems and the fabrication of custom printed circuit boards. These smart meters enable remote monitoring of gas consumption, thereby optimizing energy management and reducing operational costs. The project exemplifies the seamless integration of embedded electronics with reliable communication protocols, ensuring efficient data acquisition and real-time system oversight.

Figure 2.6: Smart Gas Meter Prototype

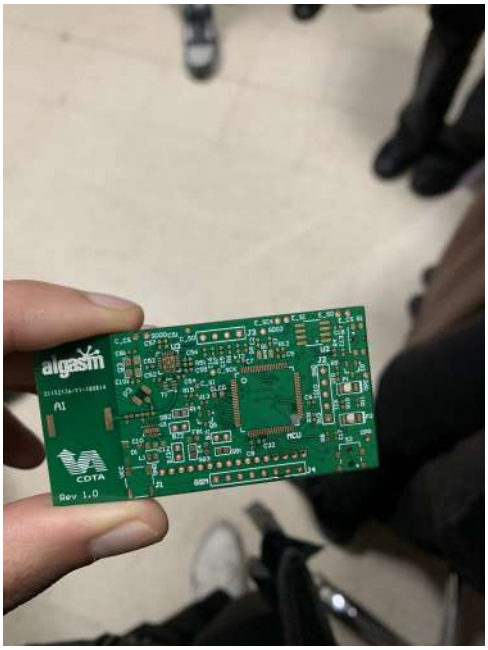

Figure 2.7: Detailed View of Gas Meter PCB

- **Drone Development and Autonomous Navigation:** The division is leading significant advances in aerial robotics by developing and testing various drone

platforms—including quadrotors and birotors—integrated with sophisticated autonomous navigation algorithms. These systems utilize advanced control techniques and sensor fusion to achieve reliable flight in dynamic environments. The project not only showcases breakthroughs in both hardware and software integration but also paves the way for versatile and robust drone applications.

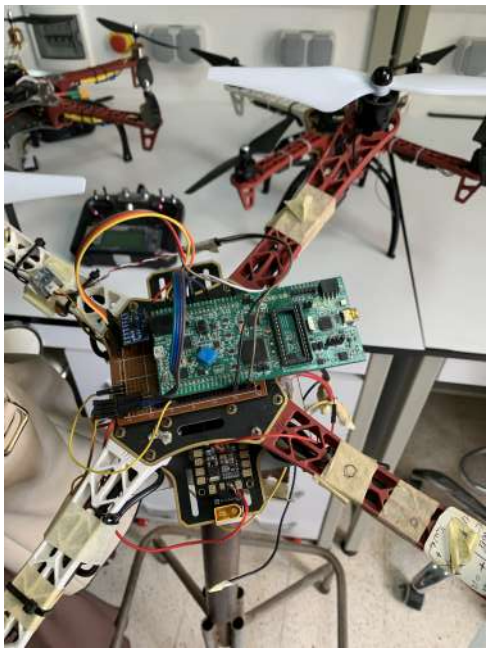

Figure 2.8: Close-up of Drone Components

Figure 2.9: All Set Drone

Figure 2.10: Autonomous Drone Platform

- **Industrial Robotics and Manufacturing Automation:** The division demonstrates innovative solutions for automating complex production processes through dedicated experimental platforms. These cutting-edge industrial robotics systems

enhance manufacturing efficiency and productivity by streamlining workflows and reducing manual intervention. The project underscores how advanced robotics and automation technologies can transform production environments.

Figure 2.11: Industrial Robotics Experimental Setup

Figure 2.12: Automation in a Manufacturing Environment

Figures 2.6 and 2.7 illustrate the smart gas meter project, while Figures 2.8–2.10 showcase various aspects of the drone development initiative. Additionally, Figures 2.11 and 2.12 provide insights into the industrial robotics setups driving automation in modern production processes.

Chapter 3

Conclusion

This report highlights the dynamic and multifaceted nature of CDTA's approach to technological innovation. Our visit underscored how the center's diverse divisions—ranging from signal processing and telecommunications to robotics and industrial automation—work in concert to address both current and future challenges in the socio-economic sector. Each division, with its specialized teams and state-of-the-art experimental setups, contributes to the creation of solutions that are not only theoretically sound but also highly practical in real-world applications.

The integration of advanced techniques such as AI-driven diagnostics in healthcare, adaptive wireless communication strategies, and robust automation systems for manufacturing demonstrates CDTA's commitment to excellence. Moreover, the active collaboration with industry partners and the seamless connection with academic institutions ensure that these innovations have a significant impact on national development while also aligning with international standards.

In summary, CDTA emerges as a vital engine for technological progress, effectively translating research into tangible products and services that enhance efficiency, safety, and quality of life. This comprehensive synergy of research, development, and practical application stands as a testament to the center's enduring vision and its strategic role in shaping the future of advanced technologies.

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